TET ASSIGNMENT

DESIGN AND DEVELOPMENT OF AN INTELLIGENT URBAN AIR QUALITY MONITORING AND HEALTH-RISK PREDICTION SYSTEM USING MACHINE LEARNING

COURSE CODE AND NAME: ITA0614 – MACHINE LEARNING

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ABSTRACT

Urban air pollution is a major environmental and public-health concern, particularly in densely populated urban regions. This project develops an intelligent machine-learning-based air-quality monitoring and health-risk prediction system using hourly air-quality observations from the Ashok Vihar monitoring station. The dataset contains particulate-matter measurements, gaseous pollutants, meteorological variables and time-related attributes.

The project follows a complete machine-learning workflow consisting of data preparation, exploratory data analysis, concept learning and version-space analysis, classification, regression, neural-network learning, genetic-algorithm-based optimization, model evaluation and an integrated warning system. The classification task is designed to identify air-quality conditions such as Good, Moderate, Poor and Hazardous using a documented target-generation methodology. PM2.5 is used as the principal pollutant indicator and as the regression target for Locally Weighted Regression.

The current development stage focuses on data preparation and exploratory analysis. The dataset contains 11,705 observations and 21 original columns. Initial quality checks found no missing values and no duplicate rows. Six exploratory visualizations have been generated to study temporal trends, distributions, outliers, correlations and hourly/monthly pollution patterns. Subsequent sections of this report will be updated with the implemented machine-learning algorithms, quantitative evaluation results, confusion matrices, model comparisons and the final warning application.

Keywords: Air Quality, PM2.5, Machine Learning, Candidate Elimination, Decision Tree, Bayesian Classification, KNN, LWR, MLP, Backpropagation, Genetic Algorithm, Health Risk Prediction

REPORT ROADMAP

- 1. Introduction
- 2. Dataset Description
- 3. System Workflow
- 4. Data Preparation and Preprocessing
- 5. Exploratory Data Analysis and Visualization
- 6. Air-Quality Target Construction
- 7. Concept Learning and Version Space
- 8. Decision Tree Classification

- 9. Bayesian Classification
- 10. K-Nearest Neighbours
- 11. Locally Weighted Regression
- 12. Multilayer Perceptron and Backpropagation
- 13. Genetic Algorithm for Feature Selection
- 14. Model Evaluation and Comparison
- 15. Integrated Air-Quality Warning System
- 16. Scalability Analysis
- 17. Bias, Variance and Uncertainty
- 18. Ethical Considerations
- 19. Limitations
- 20. Future Improvements
- 21. Sustainable Development Goal Mapping
- 22. Conclusion
- 23. References
- 24. Appendix

1. INTRODUCTION

1.1 Background

Air pollution consists of particulate matter and gaseous pollutants that can adversely affect human health and environmental quality. Urban monitoring stations continuously collect measurements such as PM2.5, PM10, nitrogen oxides, sulphur dioxide, ozone and carbon monoxide. Because pollution levels change with time, weather and emission conditions, machine learning can be used to identify patterns and support prediction and warning systems.

1.2 Problem Statement

The objective is to design an intelligent system capable of learning from historical urban air-quality observations and producing useful predictions about air-quality conditions and pollution-related risk. The system should compare multiple machine-learning approaches and provide an interpretable warning mechanism for users.

1.3 Objectives

- Prepare and analyse the Ashok Vihar hourly air-quality dataset.
- Study pollutant distributions, temporal behaviour and relationships among variables.
- Implement concept learning using Candidate Elimination and Version Space.
- Build and interpret a Decision Tree classifier.
- Implement Bayesian classification using Bayes theorem and maximum-likelihood probability estimation.
- Implement KNN from first principles and study the effect of different K values.
- Implement Locally Weighted Regression from first principles for PM2.5 prediction.
- Develop an MLP with backpropagation and experiment with model parameters.
- Use a Genetic Algorithm for feature selection/model optimization.
- Compare models using appropriate classification and regression metrics.
- Develop an integrated warning system that converts predictions into understandable recommendations.

1.4 Scope

The project covers the complete machine-learning pipeline from raw data preparation to an application-oriented warning system. The present document contains the completed data-analysis stage and a structured framework for the remaining algorithmic stages. Numerical model results will be inserted after the corresponding Colab implementations are executed and validated.

2. DATASET DESCRIPTION

The selected dataset is the Ashok Vihar hourly air-quality dataset, stored as AshokVihar_Hourly.csv. The original file contains 11,705 observations and 21 columns. One column, "Unnamed: 0", is an unnecessary CSV index field and is removed during preprocessing, leaving 20 useful attributes.

Attribute Group	Variables	Purpose
Particulate matter	PM2.5, PM10	Primary indicators of particulate pollution
Gaseous pollutants	NO, NO2, SO2, Ozone, CO, Benzene, NH3, NOx	Represent major atmospheric pollutants
Meteorological	AT, BP, SR, RH, WS, WD	Capture environmental conditions influencing pollution
Time	year, month, day, hour	Support temporal and seasonal analysis

2.1 Dataset Quality Summary

Measure	Observed Value
Original rows	11,705
Original columns	21
Useful columns after index removal	20
Missing values detected	0
Duplicate rows detected	0
PM2.5 mean	103.76
PM2.5 median	62.25
PM2.5 standard deviation	114.19
PM2.5 minimum	0.25
PM2.5 maximum	964.00

3. SYSTEM WORKFLOW

The proposed system follows a staged workflow. Raw hourly observations are first loaded and checked for data-quality issues. After preprocessing, exploratory analysis is performed to identify distributions, trends and correlations. The cleaned data are then transformed into suitable classification and regression targets. Multiple machine-learning algorithms are implemented and evaluated, followed by feature optimization and integration into a warning system.

Raw CSV Data → Cleaning → EDA → Target Construction → Concept Learning → ML Models → Optimization → Evaluation → Warning System

4. DATA PREPARATION AND PREPROCESSING

4.1 Data Loading

The CSV file was loaded into a Pandas DataFrame. Pandas provides convenient operations for inspecting, cleaning, transforming and analysing tabular data.

4.2 Missing-Value Check

A column-wise missing-value check was performed. The dataset contains zero missing values, so no imputation was required for the original observations.

4.3 Duplicate Check

Duplicate rows were checked using Pandas. No duplicate observations were detected.

4.4 Removal of Unnecessary Index

The column “Unnamed: 0” was removed because it represents a stored CSV index rather than an air-quality measurement or predictive feature.

4.5 Temporal Feature Construction

The year, month, day and hour attributes were combined into a datetime variable. The resulting observations were sorted chronologically to support temporal analysis.

4.6 Data Validation

Pollutant columns were checked for physically invalid negative values. The preprocessing pipeline also contains handling logic for invalid values so that later model training does not receive inappropriate pollutant concentrations.

5. EXPLORATORY DATA ANALYSIS AND VISUALIZATION

Exploratory data analysis was performed to understand the behaviour of PM2.5 and its relationship with other variables. Six visualizations were generated. These figures will also be used as evidence for the data-analysis and visualization component of the assignment.

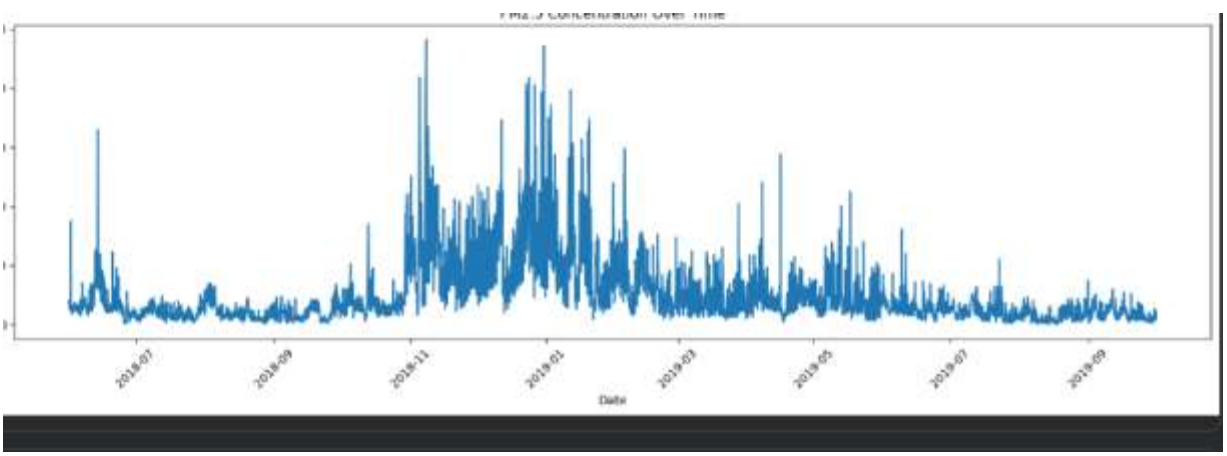

Figure 1. PM2.5 Concentration Over Time.

The time-series plot shows substantial variation in PM2.5 concentration across the observation period. Large pollution episodes are visible as sharp peaks, while extended lower-concentration periods occur between episodes. This indicates strong temporal variability and supports the need for time-aware analysis.

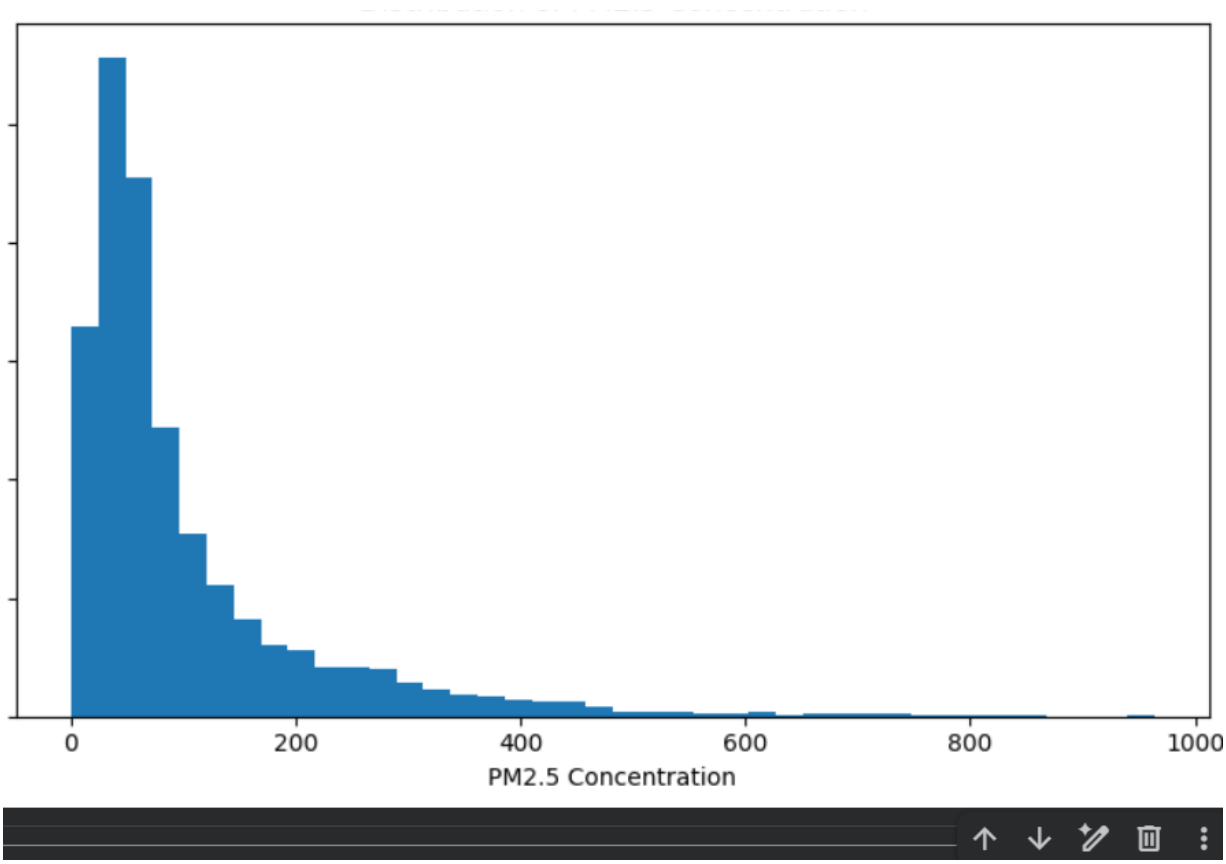

Figure 2. Distribution of PM2.5 Concentration.

The PM2.5 distribution is strongly right-skewed. Most observations are concentrated at relatively lower concentrations, while a smaller number of observations extend to very high values. This behaviour is important when selecting scaling, regression and classification strategies.

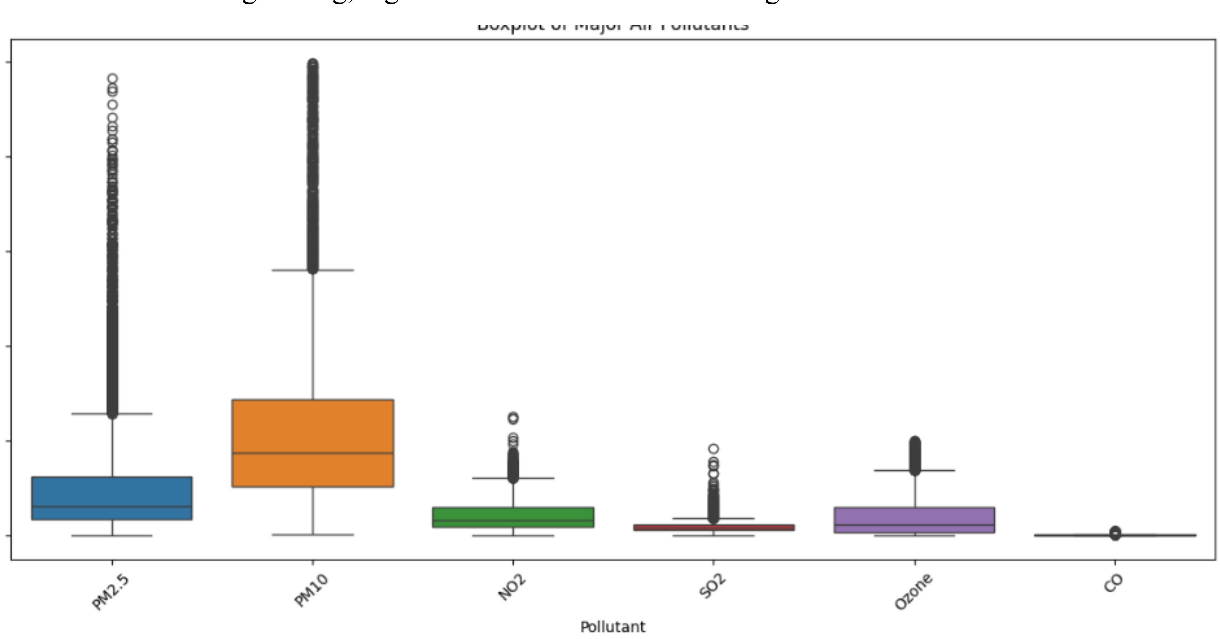

Figure 3. Boxplot of Major Air Pollutants.

The boxplots show differences in central tendency and spread among the major pollutants. Numerous high-value observations appear as outliers, particularly for PM2.5 and PM10. These observations should not automatically be deleted because extreme pollution episodes may represent genuine environmental conditions.

Figure 4. Correlation Matrix of Air-Quality Variables.

The correlation matrix shows the strength and direction of linear relationships among numerical variables. PM2.5 has a strong positive relationship with PM10 and noticeable relationships with several gaseous pollutants. Meteorological variables also show meaningful relationships with pollution variables, supporting their use as predictive features.

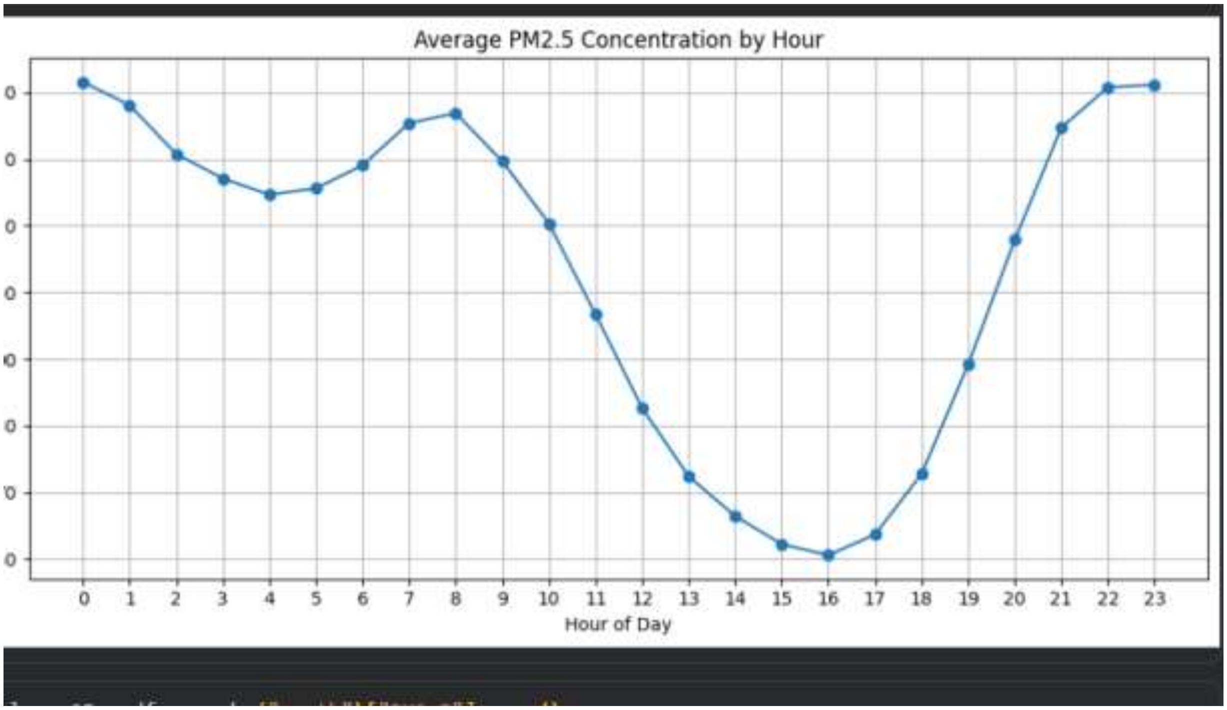

Figure 5. Average PM2.5 Concentration by Hour of Day.

The hourly pattern shows that average PM2.5 is higher during late-night and early-morning hours, decreases through the afternoon, and rises again during evening and late-night hours. This indicates a strong diurnal pattern and demonstrates why the hour attribute can be informative for prediction.

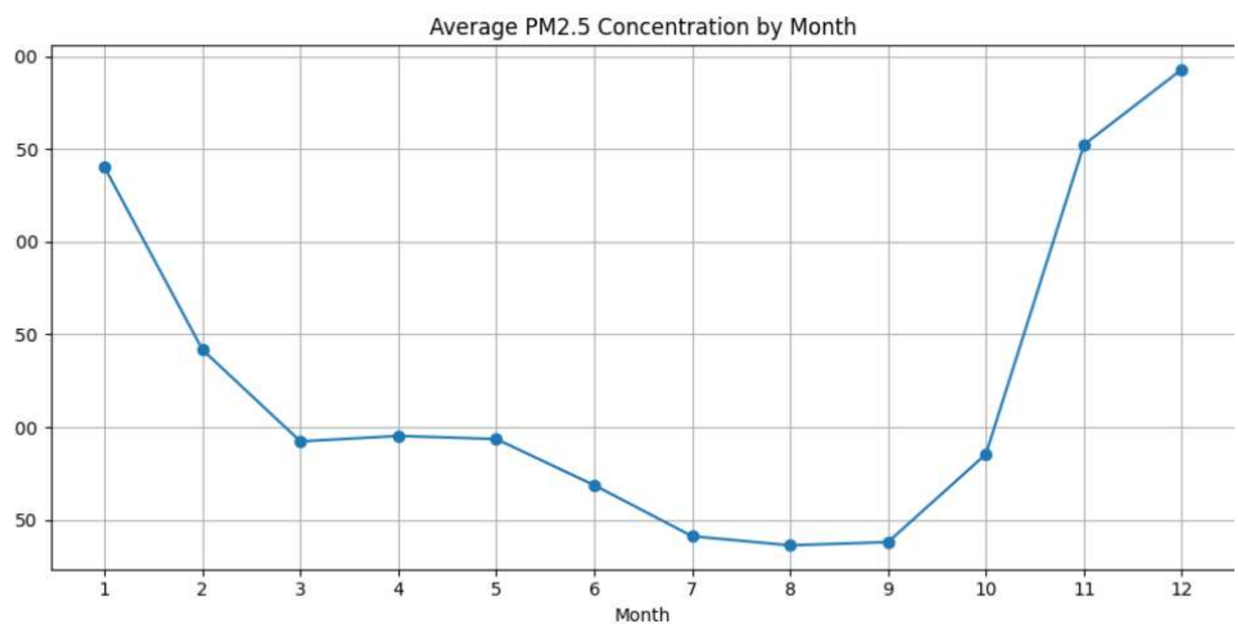

Figure 6. Average PM2.5 Concentration by Month.

The monthly analysis shows strong seasonal variation. Average PM2.5 is comparatively high in January, rises sharply again during November and December, and is much lower during the middle months of the year. This suggests that seasonal and meteorological conditions have a major influence on observed pollution levels.

5.1 Key EDA Findings

- PM2.5 exhibits strong temporal variability with several high-pollution episodes.
- The PM2.5 distribution is highly right-skewed, with a long upper tail.
- The pollutant boxplots indicate substantial variation and many high-value observations.
- PM2.5 and PM10 show a strong positive linear association in the correlation analysis.
- Average PM2.5 changes considerably across hours of the day.
- Average PM2.5 also changes considerably across months, indicating seasonal behaviour.

6. AIR-QUALITY TARGET CONSTRUCTION

The original dataset does not contain a ready-made AQI_Category field. Therefore, the classification target must be derived using a clearly documented and scientifically justified threshold methodology rather than assigning categories arbitrarily. The final implementation will define a reproducible mapping from pollutant/AQI-related measurements to four categories: Good, Moderate, Poor and Hazardous.

6.1 Planned Classification Target

Category	Meaning
Good	Low pollution / relatively low health concern
Moderate	Noticeable pollution / sensitive groups may be affected
Poor	High pollution / increased health concern
Hazardous	Very high pollution / serious health concern

The exact numeric thresholds and the pollutant-to-index calculation will be documented in the final implementation section after the target-generation method is finalized and validated.

7. CONCEPT LEARNING AND VERSION SPACE

This section will formulate air-quality classification as a concept-learning problem. Candidate hypotheses will be represented using selected categorical/thresholded features. The target concept will be the desired air-quality class. Candidate Elimination will maintain the Specific Boundary (S) and General Boundary (G), and the Version Space will contain all hypotheses consistent with the training examples.

7.1 Inductive Bias

The implementation will assume that the chosen pollutant and environmental features are sufficient to distinguish the target classes and that the training examples are representative of the underlying air-quality concept.

8. DECISION TREE CLASSIFICATION

A Decision Tree classifier will be developed for the four air-quality classes. The tree will select features using an impurity measure such as Gini impurity or Information Gain. The final tree will be visualized and interpreted so that the most influential decision variables and class-separation rules can be explained.

8.1 Required Output

- Decision-tree structure/visualization.
- Training and test performance.
- Confusion matrix.

- Interpretation of important splits.

9. BAYESIAN CLASSIFICATION

The Bayesian classifier will estimate the posterior probability of each air-quality category using Bayes theorem. Maximum-likelihood estimation will be used to estimate the required model parameters. The class with the highest posterior probability will be selected as the prediction.

9.1 Bayes Theorem

$P(C|X) = [P(X|C) \times P(C)] / P(X)$, where C represents a class and X represents the observed feature vector.

The final report will include an example probability calculation using actual or representative dataset values.

10. K-NEAREST NEIGHBOURS

KNN will be implemented from first principles rather than relying only on a pre-built KNN classifier. The implementation will calculate distances between a test sample and training samples, sort the distances, select the K nearest neighbours and determine the predicted class using majority voting.

10.1 Effect of K

Several K values will be tested. A validation curve will be generated to show how classification performance changes as K increases. Small K values may be sensitive to noise, while excessively large K values may oversmooth local class structure.

11. LOCALLY WEIGHTED REGRESSION

Locally Weighted Regression will be implemented from first principles to predict PM2.5. Nearby training observations will receive larger weights than distant observations. A weighting parameter such as tau will control how rapidly the influence decreases with distance.

11.1 Regression Comparison

LWR will be compared with a conventional regression baseline using MAE and RMSE. The effect of different weighting parameters will also be analysed.

12. MULTILAYER PERCEPTRON AND BACKPROPAGATION

A Multilayer Perceptron will be developed for air-quality classification. The network will include an input layer, one or more hidden layers and an output layer. During forward propagation, activations will be computed through the network. Backpropagation will calculate gradients of the loss with respect to the weights, and gradient descent will update the parameters.

12.1 Parameter Experiments

- Number of hidden neurons
- Learning rate
- Number of epochs
- Activation function

- Batch size

The final implementation will report how these parameters affect learning behaviour and validation performance.

13. GENETIC ALGORITHM FOR FEATURE SELECTION

A Genetic Algorithm will be used to search for useful subsets of input features. Each chromosome will represent a candidate feature subset using a binary encoding, where 1 means the feature is selected and 0 means it is excluded.

13.1 Genetic Operations

- Fitness evaluation
- Selection
- Crossover
- Mutation
- Replacement / generation update

Fitness will be based on predictive performance while considering the number of selected features. The optimized feature subset will be compared with the original feature set.

14. MODEL EVALUATION AND COMPARISON

The final evaluation stage will compare the developed models using metrics appropriate to each task.

Metric	Formula / Meaning	Used For
Accuracy	Correct predictions / total predictions	Classification
Precision	TP / (TP + FP)	Classification
Recall	TP / (TP + FN)	Classification
F1-score	2 × Precision × Recall / (Precision + Recall)	Classification
MAE	Mean absolute prediction error	Regression
RMSE	Square root of mean squared prediction error	Regression

14.1 Planned Comparison Outputs

- Confusion matrix for each classification model.
- Accuracy, precision, recall and F1-score comparison.
- KNN validation curve.
- LWR prediction and error plots.
- MLP training/validation performance.
- Genetic Algorithm fitness progression.
- Overall model-comparison chart.

15. INTEGRATED AIR-QUALITY WARNING SYSTEM

The final application will convert model predictions into understandable air-quality warnings. The user interface will accept relevant pollutant/environmental inputs or process a selected observation and display the predicted air-quality category, pollution indicators and a health-oriented recommendation.

15.1 Example Warning Logic

- Good → Normal outdoor activity is generally appropriate.
- Moderate → Sensitive individuals should consider reducing prolonged exposure.
- Poor → Reduce prolonged outdoor activity and monitor symptoms.
- Hazardous → Avoid unnecessary outdoor exposure and follow local health guidance.

These messages are informational and should not be presented as a medical diagnosis.

16. SCALABILITY ANALYSIS

Scalability will be evaluated by running selected algorithms on progressively larger portions of the dataset, for example 20%, 40%, 60%, 80% and 100%. Training/runtime and predictive performance will be recorded. This will demonstrate how computational cost changes as the number of observations increases.

17. BIAS, VARIANCE AND UNCERTAINTY

The final report will discuss the trade-off between bias and variance for the selected models. KNN with very small K may have high variance, while overly large K may have higher bias. Decision Trees may overfit without suitable depth control. Neural networks may also overfit if the model is too complex relative to the training data. Prediction uncertainty will be acknowledged, particularly during extreme pollution events or observations outside the range well represented in training data.

18. ETHICAL CONSIDERATIONS

- Air-quality predictions can influence health-related decisions, so uncertainty should be communicated clearly.
- The system should not be presented as a medical diagnostic tool.
- Model bias may occur if the training data are not representative of other locations or time periods.
- False alarms and missed hazardous conditions both have potential consequences and should be evaluated.
- The system should avoid overstating prediction confidence.

19. LIMITATIONS

- The dataset represents a specific monitoring location and time period, so generalization to other cities or stations may be limited.
- The original dataset does not directly provide an AQI category, requiring a documented target-construction procedure.
- Extreme pollution values can make some models sensitive to outliers.
- Machine-learning performance depends on the quality and representativeness of historical observations.
- Real-time deployment would require a continuous and reliable source of current sensor measurements.

20. FUTURE IMPROVEMENTS

- Integrate live air-quality sensor or API data.
- Add additional monitoring stations for spatial prediction.
- Use time-series models for multi-hour forecasting.
- Perform systematic hyperparameter optimization.

- Add uncertainty estimation and confidence-aware warnings.
- Deploy the system as a web or mobile application.
- Continuously retrain the model as new observations become available.

21. SUSTAINABLE DEVELOPMENT GOAL MAPPING

SDG	Project Contribution
SDG 3 – Good Health and Well-Being	Supports awareness of pollution-related health risk.
SDG 9 – Industry, Innovation and Infrastructure	Demonstrates intelligent monitoring and machine-learning infrastructure.
SDG 11 – Sustainable Cities and Communities	Supports cleaner and safer urban environments.
SDG 13 – Climate Action	Encourages data-driven environmental monitoring and awareness.

22. CONCLUSION

The Ashok Vihar hourly air-quality dataset provides a strong foundation for developing an intelligent urban air-quality monitoring and prediction system. Initial preprocessing confirms that the dataset is structurally suitable for machine-learning analysis, with no missing values or duplicate rows detected. Exploratory analysis reveals strong temporal and seasonal variation in PM2.5, a highly right-skewed distribution and meaningful relationships among pollutants and environmental variables.

The remaining project stages will build on this foundation by implementing concept learning, classical machine-learning classifiers, regression, neural-network learning and genetic optimization. Their performance will then be compared and integrated into a practical warning system. The final system will emphasize both predictive performance and responsible communication of uncertainty and health-related risk.

23. REFERENCES

- Mitchell, T. M. Machine Learning. McGraw-Hill.
- Han, J., Kamber, M., and Pei, J. Data Mining: Concepts and Techniques. Morgan Kaufmann.
- Russell, S. and Norvig, P. Artificial Intelligence: A Modern Approach. Pearson.
- Course materials and laboratory notes for ITA0614 – Machine Learning.
- Air-quality dataset: AshokVihar_Hourly.csv, used as the primary project dataset.

24. APPENDIX

Appendix A – Current Colab Development Status

Completed in the current development stage:

- Dataset upload and loading.
- Dataset shape and column inspection.
- Missing-value analysis.
- Duplicate-row analysis.
- Removal of the unnecessary index column.
- Statistical summary.
- PM2.5 mean, median, standard deviation, minimum and maximum.

- Pollutant validity check.
- Datetime construction and chronological sorting.
- Six exploratory visualizations.

Appendix B – Figures Included in This Report

- Figure 1. PM2.5 Concentration Over Time.
- Figure 2. Distribution of PM2.5 Concentration.
- Figure 3. Boxplot of Major Air Pollutants.
- Figure 4. Correlation Matrix of Air-Quality Variables.
- Figure 5. Average PM2.5 Concentration by Hour of Day.
- Figure 6. Average PM2.5 Concentration by Month.

Appendix C – Results to Be Added After Model Implementation

- Candidate Elimination / Version Space output.
- Decision Tree results and confusion matrix.
- Bayesian classifier results and probability example.
- KNN results for multiple K values and validation curve.
- LWR results for multiple weighting parameters.
- MLP/backpropagation training and evaluation results.
- Genetic Algorithm feature-selection results.
- Complete model-comparison table.
- Scalability measurements.
- Final warning-system screenshots.